



CSCE 633-700: Machine Learning

Linear Perceptron Algorithm

Outline

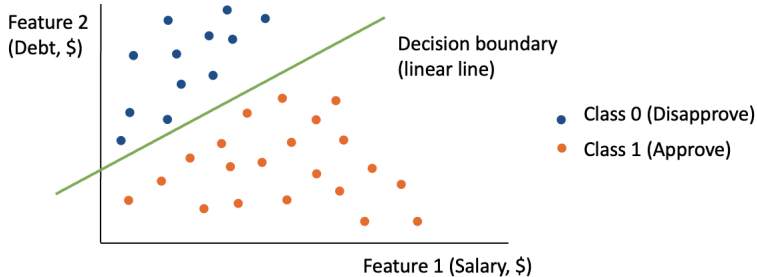
- Perceptron
- Example & Representation
- Perceptron Learning Algorithm (PLA)

(* Part of the following slides are taken from Dr. Malik Magdon-Ismail's Machine Learning class.)

Perceptron: Example

Credit approval or denial

- Task: Approve or deny credit (binary classification task)
- Features: Salary, debt, years in residence, etc.



CREDIT APPROVED
Today!!

Perceptron: Example & Representation

- Input vector $\mathbf{x} = [x_1, \dots, x_d]^T$
- Assign importance to input features and compute a Credit Score

$$CreditScore = \sum_{i=1}^D w_i x_i$$

In the above, **weights convey importance if features are in the same range**

- Approve credit if $\sum_{i=1}^D w_i x_i > threshold$
- Deny credit if $\sum_{i=1}^D w_i x_i < threshold$
- How to choose the importance of weights?

Perceptron: Example & Representation

- Approve credit if $\sum_{i=1}^D w_i x_i > \text{threshold}$
- Deny credit if $\sum_{i=1}^D w_i x_i < \text{threshold}$
- Can be written more formally as

$$h(\mathbf{x}) = \text{sign} \left(\sum_{i=1}^d (w_i x_i) + w_0 \right)$$

or

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x})$$

where $\mathbf{w} = [w_0, w_1, \dots, w_d]^T$ and $\mathbf{x} = [1, x_1, \dots, x_d]^T$

Perceptron: Representation

Classify data into two classes

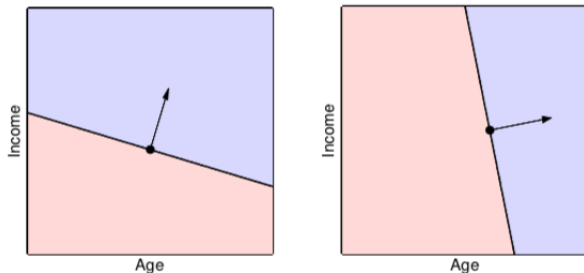
- **Input:** $\mathbf{x} \in \mathbb{R}^D$ (features, attributes, etc.)
- **Output:** $y \in \{-1, 1\}$ (labels)
- **Model:** $h : \mathbf{x} \rightarrow y$

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x}) = \begin{cases} 1, & \mathbf{w}^T \mathbf{x} > 0 \\ -1, & \mathbf{w}^T \mathbf{x} < 0 \end{cases}$$

Perceptron: Representation

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x})$$

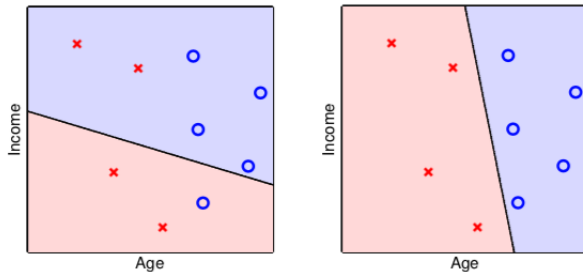
(Problem 1.2 in LFD)



How to find the perpendicular vector to a line?

The vector perpendicular to line $w_1x_1 + w_2x_2 + w_0 = 0$ can be written as $\mathbf{n} = [w_1, w_2]^T$ or $\mathbf{n} = [w_0, w_1, w_2]^T$ (if the line does not pass through the beginning of the axis) (See Supplementary Handout)

Perceptron: Representation



A perceptron fits the data by using a line to separate the $+1$ from -1 data.

Fitting the data: How to find a hyperplane that *separates* the data?

(“It’s obvious - just look at the data and draw the line,” is not a valid solution.)

Perceptron Learning Algorithm

Idea! Start with some weight vector and try to improve it.

